FLAGELLATES

The whip-like propulsion of single-celled organisms has evolved in several groups of protoctists, some unrelated. But it predominates in the Excavata.

Flagellates are powerful swimming microbes that move by the whip-like action of one or more external threads called flagella. Many are predators, preying on smaller organisms such as bacteria. However, unlike changeable amoebas, these flagellates have a fixed, rigid shape and direct food into a cellular "mouth" at the base of the flagella. Remarkably, however, some, such as *Euglena*, have a degree of behavioural versatility that provides them with plant- and animal-type nutrition, depending upon circumstance. When exposed to bright light these organisms can photosynthesize; but in darkness their light-absorbing cell organelles, known as chloroplasts, shrivel and they revert to being a predator.

LIVING INSIDE ANIMALS

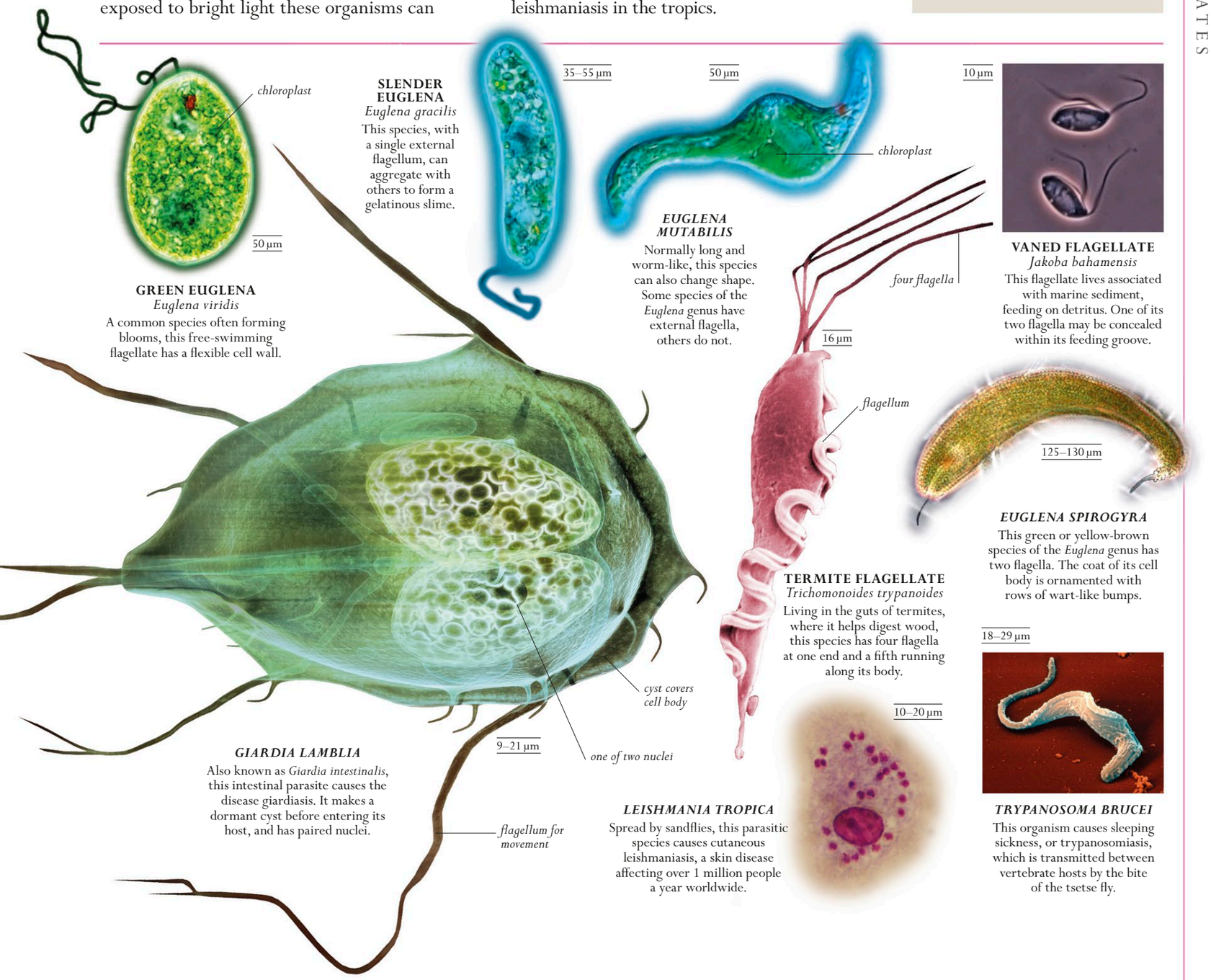
Many flagellates of this group lack the ordinary cellular features that would enable them to use oxygen in their respiration and survive inside the oxygen-poor environment of animal guts. Many are extraordinarily specialized, subsisting on the partially-digested food inside the abdomens of insects, but causing their host no harm. Others are parasites that cause devastating disease, including in humans. One notorious group, the trypanosomes, are transmitted by biting insects and are responsible for sleeping sickness and leishmaniasis in the tropics.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
CLADE	EXCAVATA
FAMILIES	40
SPECIES	About 2,500

DEBATE

EVOLUTION DOWN-SIZING

The flagellates inside termite guts lack mitochondria, the cell organelles most other protoctists use to respire with oxygen. Some think these flagellates are primitive protoctists. Others say they are more advanced organisms, and lost their mitochondria by evolving in an oxygen-poor environment.



GREEN EUGLENA
Euglena viridis
A common species often forming blooms, this free-swimming flagellate has a flexible cell wall.

SLENDER EUGLENA
Euglena gracilis
This species, with a single external flagellum, can aggregate with others to form a gelatinous slime.

EUGLENA MUTABILIS
Normally long and worm-like, this species can also change shape. Some species of the *Euglena* genus have external flagella, others do not.

VANED FLAGELLATE
Jakoba bahamensis
This flagellate lives associated with marine sediment, feeding on detritus. One of its two flagella may be concealed within its feeding groove.

EUGLENA SPIROGYRA
This green or yellow-brown species of the *Euglena* genus has two flagella. The coat of its cell body is ornamented with rows of wart-like bumps.

TERMITE FLAGELLATE
Trichomonoides trypanoides
Living in the guts of termites, where it helps digest wood, this species has four flagella at one end and a fifth running along its body.

GIARDIA LAMBLIA
Also known as *Giardia intestinalis*, this intestinal parasite causes the disease giardiasis. It makes a dormant cyst before entering its host, and has paired nuclei.

LEISHMANIA TROPICA
Spread by sandflies, this parasitic species causes cutaneous leishmaniasis, a skin disease affecting over 1 million people a year worldwide.

TRYPANOSOMA BRUCEI
This organism causes sleeping sickness, or trypanosomiasis, which is transmitted between vertebrate hosts by the bite of the tsetse fly.